

Jiageng Liu

• Phone: +1-4132654677 • Email: jiagengliu@umass.edu • Page: <https://jiagengliu02.github.io>

EDUCATION

University of Massachusetts Amherst	Massachusetts, United States
<i>Master in Computer Science</i> GPA: 4.00/4.00	09/2024-01/2026
Zhejiang University	Hangzhou, China
<i>B.E. in Artificial Intelligence</i> GPA: 3.75/4.00 Major GPA (Third year): 3.96/4.00	09/2020-07/2024
<i>Courses:</i> Natural Language Processing (97), Machine Learning (92), Computer Vision (90), Design of AI Product (96)	

PUBLICATIONS

- Yuncong Yang*, **Jiageng Liu***, Zheyuan Zhang et al. Test-Time Scaling with World Models for Spatial Reasoning. *Advances in Neural Information Processing Systems (NeurIPS2025)*.
[Program Website](#)
- Yian Wang*, Xiaowen Qiu*, **Jiageng Liu*** et al. Architect: Generating Vivid and Interactive 3D Scenes with Hierarchical 2D Inpainting. *Advances in Neural Information Processing Systems (NeurIPS2024)*.
[Program Website](#)
- Zeyuan Yang*, **Jiageng Liu***, Peihao Chen, Anoop Cherian, Tim K. Marks, Jonathan Le Roux, Chuang Gan. RILA: Reflective and Imaginative Language Agent for Zero-Shot Semantic Audio-Visual Navigation. *Advances in Computer Vision and Pattern Recognition (CVPR2024)*.
[Program Website](#)
- Wang Zehan, Yang Zhao, Xize Cheng, Haifeng Huang, **Jiageng Liu** et al. Connecting multi-modal contrastive representations. *Advances in Neural Information Processing Systems (NeurIPS2023)*.
[Program Website](#)
- Yuhang Xu, Wanxu Xia, Yipeng Chen, **Jiageng Liu** et al. WordDiffuser: Helping beginners cognize and memorize glyph-semantic pairs using diffusion. *In Proceedings of the IEEE International Symposium on Computational Intelligence and Design (ISCID2023)*.
[Program Website](#)

RESEARCH EXPERIENCES

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| Large Reconstruction Model for Stylized 3DGS Reconstruction | University of California, Merced, United States |
| <i>Research Leader, Advisor:</i> Ming-Hsuan Yang | 06/2025-10/2025 |
- Most style-transfer methods for 3D Gaussian Splatting are optimization-based, limiting both speed and precision.
 - Large reconstruction models trained on edited multi-view images often suffer cross-view inconsistencies; multi-view 3D reconstruction also incurs heavy VRAM usage.
 - Built on NoPoSplat, our framework replaces test-time optimization with a large feed-forward reconstruction model, dramatically accelerating 3DGS style-transfer reconstruction while producing consistent paired multi-view training data and reducing VRAM footprint.
 - Experiments show substantial speedups and accuracy gains over prior work.
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| Test-Time Scaling with World Models for Spatial Reasoning (NeurIPS2025 accepted) | Microsoft, United States |
| <i>Research Leader, Advisor:</i> Jianwei Yang | 02/2025-06/2025 |
- State-of-the-art VLMs often fail on 3D spatial reasoning (e.g., perspective shifts). We address this by pairing a VLM with a controllable video world model that imagines egocentric views from a single image.
 - Proposed test-time scaling via spatial beam search: the VLM sketches camera trajectories; the world model paints corresponding views; the VLM stores helpful information as multi-view evidence step-by-step to analyze the answers.
 - On the SAT benchmark, Our method works with four different VLMs (GPT-4o/4.1, InternVL3-14B, o1) and two world models (our SWM and Stable-Virtual-Camera), and can even surpass RL-scaled baselines when combined.
 - Co-authored the paper that was accepted as poster in NeurIPS2025.

Virtual Community for Scalable 3D Scene and Embodied Agents (ICLR2026 under review) Umass, United States

Research Participant, Advisor: [Chuang Gan](#)

07/2024-10/2024

- The Motivation of Virtual Community is to feature large-scale community scenarios derived from the real world and construct a social world simulation platform designed to support embodied AI research.
- Scalable 3D Scene creation: which supports the generation of expansive outdoor and indoor environments at any location and scale, addressing the lack of a large-scale, interactive, open world scene for embodied AI research.
- Embodied agents with grounded characters and social relationship networks: the first to simulate socially connected agents at a community level, that also have scene-grounded characters.
- I take responsibility of multi-room and multi-scene indoor scene automatic generation, which based on Architect.
- Co-authored the paper that was under review in ICLR2024.

Lifting 2D Inpaint to Interactive 3D Scene Generation (NeurIPS2024 accepted) MIT-IBM Watson AI lab, United States

Research Leader, Advisor: [Chuang Gan](#)

11/2023-06/2024

- Aiming at create a general framework to generate sufficient and delicate interactive 3D scenes for Robotics and Embodied AI research, such as RL training.
- Created a framework name Architect, which utilizes pre-trained 2D image generative models to inpaint, and visual foundation model for perception to lift 2D image into 3D scene. Implement the inpainting iteratively and hierarchically.
- Architect is capable to generate infinity context-rich, fine-grind, interactive, indoor scenes.
- Co-authored the paper that was accepted as poster in NeurIPS2024.

LLM Commonsense Assisted Navigation (CVPR2024 accepted)

MIT-IBM Watson AI lab, United States

Research Leader, Advisor: [Chuang Gan](#)

06/2023-11/2023

- Researched the spatial reasoning and multi-modal information processing capabilities of LLM.
- Utilized a zero-shot framework on Semantic Audio-Visual Navigation (SAVN) embodied AI task in simulation environment Habitat. Integrated LLM to imagine room layout, give high-level guidance into our framework Reflective and Imaginative Language Agent (RILA).
- Achieved results surpassed previous baselines (SOTA) that requires training, using LLM's commonsense.
- Co-authored the paper that was accepted as poster in CVPR2024.

Connecting Multi-modal Contrastive Representations (NeurIPS2023 accepted)

Zhejiang University, China

Research Participant, Advisor: [Zhou Zhao](#)

02/2023-05/2023

- Conducted research on Multi-modal Contrastive Representation (MCR) learning, which aims to encode different modalities into a semantically aligned shared space.
- Devised a novel training-efficient method Connecting-MCR (C-MCR) for learning MCR without paired data.
- Designed a framework of connecting frozen CLIP and CLAP using our C-MCR to train an audio-visual model, dataset comes from six cross-modal. Analyzed the model on zero-shot downstream tasks comparing with SOTA methods.
- Wrote a paper that was published to NeurIPS2023 with the title Connecting Multi-modal Contrastive Representations.

HONORS & AWARDS

National Scholarship (Top 0.2%)	Ministry of Education, China
Top ten raising star in College of CS	College of Computer Science & Technology, Zhejiang University, China
First-class Scholarship (Top 3%)	Zhejiang University, China
Learnings' Innovation Scholarship (Top 1.5%)	College of Computer Science & Technology, Zhejiang University, China
First-class of Basic Top Subject Scholarship (Top 10%)	Zhejiang University, China
Second prize in Chinese High School Mathematics League	Chongqing, China

SKILLS & INTERESTS

Research Interest: Embody AI, Computer Vision, 3D Generation, World Model

Skills: Python(mainly, pytorch, numpy, tensorflow), C++, C#, HTML, CSS, JavaScript, SQL, MATLAB

Hobby: Jogging, Cooking, Piano